

Project Design Phase
Solution Architecture

Date	01 October 2026
Team ID	6ab7f80ad083b0e2ee72b17e
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

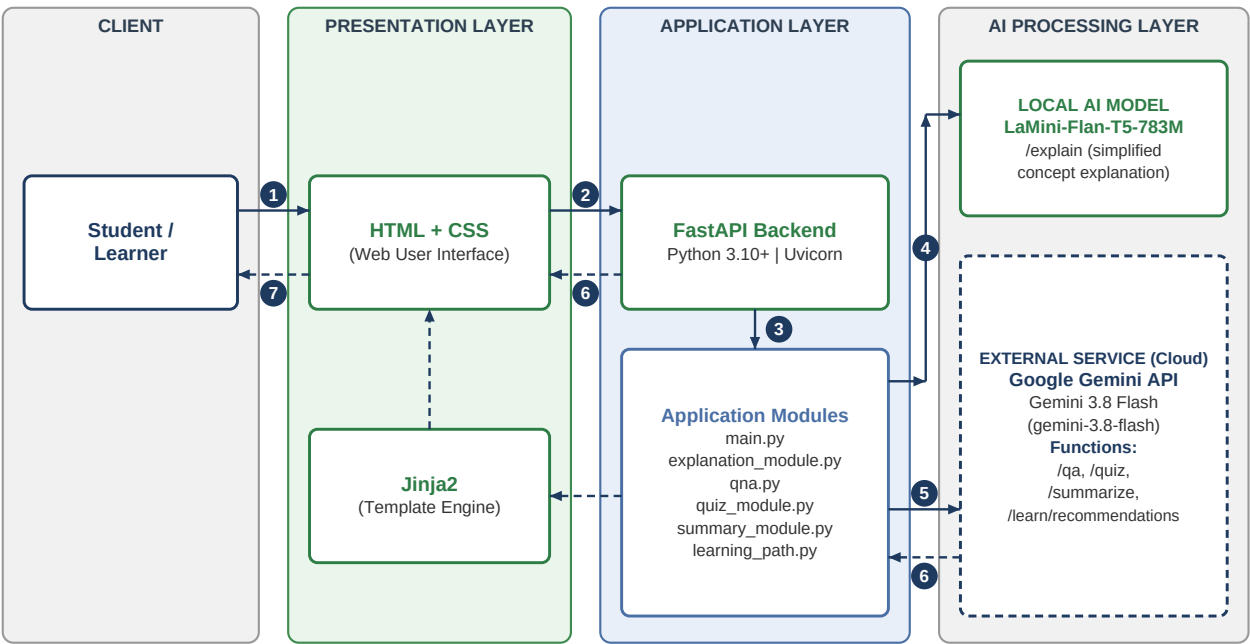


Figure 1: Architecture and data flow of the EduGenie learning assistant prototype

Data flow: 1 Student submits a request in the web interface. 2 The UI forwards it to the FastAPI backend. 3 FastAPI routes it to the matching feature module. 4 /explain is handled by the local LaMini-Flan-T5-783M model. 5 /qa, /quiz, /summarize and /learn/recommendations are sent to the Gemini API (gemini-3.8-flash). 6 The AI output returns to FastAPI. 7 The response is rendered through Jinja2 and HTML and shown to the student. No database or cloud storage is used in the prototype.

Reference: EduGenie Project Design Phase-II: Technology Stack (Architecture & Stack).